

ROTH, Germany

WMO: 107650

Lat: 49.2162N

Lon: 11.1035E

Elev: 387

StdP: 96.76

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-10.0	-14.7	1.1	-12.3	-11.8	1.4	-8.5	10.1	6.6	8.8	4.2	1.2	130	0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.0	31.2	19.3	29.0	18.4	27.1	17.8	20.2	28.2	19.5	26.8	18.6	25.3	2.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.9	13.4	21.8	17.0	12.7	21.4	16.1	12.0	20.9	59.7	28.6	56.9	26.7	54.2	25.3	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.4	5.4	-17.1	34.3	3.3	2.0	-19.5	35.7	-21.5	36.9	-23.4	38.0	-25.8	39.4		
(5)			-17.3	21.9	3.3	1.0	-19.7	22.6	-21.6	23.2	-23.5	23.8	-25.9	24.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.0	-0.4	0.8	4.5	8.8	13.5	16.8	18.5	18.2	13.7	9.2	4.0	0.0	(6)
(7)		DBStd	7.91	4.85	4.80	4.23	4.00	3.78	3.78	3.32	3.15	3.41	3.93	3.74	4.39	(7)
(8)		HDD10.0	1395	324	257	176	68	11	1	0	0	7	63	181	309	(8)
(9)		HDD18.3	3520	582	490	430	285	157	74	39	40	143	283	429	567	(9)
(10)		CDD10.0	1037	0	0	4	33	118	203	265	255	117	38	2	0	(10)
(11)		CDD18.3	120	0	0	0	0	6	27	45	37	4	1	0	0	(11)
(12)		CDH23.3	1619	0	0	0	12	127	367	559	483	69	3	0	0	(12)
(13)		CDH26.7	507	0	0	0	1	25	108	198	164	9	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.4	2.5	2.6	2.3	2.3	2.1	2.2	1.9	1.9	1.9	2.1	2.3	(14)
(15)	Precipitation	PrecAvg	743	53	47	52	44	70	81	87	74	57	57	61	60	(15)
(16)		PrecMax	1055	113	123	154	111	146	136	140	155	136	176	150	149	(16)
(17)		PrecMin	551	5	11	8	10	10	28	23	27	10	12	0	12	(17)
(18)		PrecStd	111	28	26	33	22	34	30	30	31	29	33	29	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.8	14.4	19.1	24.9	29.2	32.5	33.7	34.8	28.1	22.7	15.2	10.9	(19)
(20)			MCWB	8.3	8.8	10.8	13.6	17.7	19.7	20.0	19.7	18.1	15.8	11.4	8.2	(20)
(21)		2%	DB	9.9	11.4	16.2	22.3	26.7	29.8	31.2	30.4	25.4	19.7	12.9	9.3	(21)
(22)			MCWB	7.7	7.3	9.7	12.8	16.5	19.0	19.4	19.0	17.3	14.6	9.9	7.6	(22)
(23)		5%	DB	8.0	9.4	13.7	19.8	24.3	27.5	28.9	28.1	23.1	17.2	11.1	7.7	(23)
(24)			MCWB	6.0	6.6	8.6	12.0	15.4	17.9	18.6	18.2	16.1	13.2	9.1	6.2	(24)
(25)		10%	DB	6.1	7.5	11.2	17.2	21.9	25.3	26.5	26.0	20.9	15.3	9.5	6.0	(25)
(26)			MCWB	4.7	5.4	7.6	10.5	14.2	17.1	17.9	17.7	15.1	12.2	7.8	5.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.3	9.2	11.7	14.9	18.7	21.3	21.2	21.1	19.4	16.7	12.0	9.2	(27)
(28)			MCDWB	11.1	12.9	16.3	22.5	26.7	28.3	29.8	29.4	25.1	21.5	14.8	10.0	(28)
(29)		2%	WB	7.7	8.0	10.4	13.4	17.3	19.9	20.2	20.0	17.9	15.1	10.5	7.7	(29)
(30)			MCDWB	9.6	10.6	14.8	20.7	25.2	27.6	28.3	28.2	23.8	18.6	12.3	9.1	(30)
(31)		5%	WB	6.1	6.8	9.2	12.2	16.1	18.8	19.4	19.1	16.8	13.8	9.3	6.3	(31)
(32)			MCDWB	7.7	9.0	12.8	18.6	22.9	25.7	26.6	26.4	21.8	16.8	10.9	7.4	(32)
(33)		10%	WB	4.7	5.5	7.9	11.1	14.9	17.7	18.5	18.2	15.6	12.6	7.9	4.9	(33)
(34)			MCDWB	6.0	7.4	10.8	16.7	20.7	24.0	24.9	24.4	19.9	14.9	9.4	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.1	7.8	9.6	12.1	12.2	11.9	12.0	12.1	11.2	9.4	6.1	5.3	(35)
(36)			MCDBR	7.7	10.7	14.6	16.9	16.9	16.6	16.9	16.9	15.4	12.2	8.9	6.8	(36)
(37)			MCWBR	5.9	7.2	8.7	8.7	7.9	7.3	6.6	7.2	8.0	7.6	6.3	5.4	(37)
(38)		5% WB	MCDBR	7.2	9.4	12.6	15.3	14.9	14.7	14.2	14.5	13.2	10.9	8.2	6.2	(38)
(39)			MCWBR	5.8	6.7	7.8	8.2	7.3	6.8	6.0	6.6	7.1	6.9	6.1	5.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.303	0.326	0.361	0.393	0.401	0.404	0.411	0.405	0.377	0.353	0.327	0.300	(40)	
(41)		taud	2.435	2.380	2.325	2.267	2.274	2.295	2.270	2.308	2.364	2.423	2.457	2.437	(41)	
(42)		Ebn at Noon	757	820	845	852	859	857	844	831	818	776	719	708	(42)	
(43)		Edh at Noon	63	84	105	123	128	126	127	117	100	80	62	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.89	1.72	2.83	4.27	5.03	5.59	5.30	4.61	3.34	2.00	0.99	0.70	(44)	
(45)		RadStd	0.14	0.28	0.36	0.56	0.57	0.46	0.50	0.44	0.43	0.24	0.12	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)		N/A	N/A	N/A	N/A	N/A	+0.39	N/A	N/A	+50	+18

Nomenclature: See separate page